HW 07 – Services

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Pseudocode:

resetworld.m

```
FUNCTION reset the robot simulation world
    GET robot's handle
    INITIALIZE message for robot's world reset service
    CALL robot's world reset service with the reset message
```

getModels.m

```
FUNCTION return an object with all the models in the simulation
    GET robot's handle
    INITIALIZE message for robot's model retrieving service
    CALL robot's model retrieving service with the model message to retrieve an object with all
        models in the simulation
    RETURN the object containing all models
```

get_model_pose.m

```
FUNCTION return an object with the current state of a model given an input model name
    GET robot's handle
    INITIALIZE message for robot's model state retrieving service
    POPULATE the message with the input model name
    TRY
        CALL robot's model state retrieving service with the message to retrieve an object with
            the current model state
        RETURN the object with the current model state
    CATCH
        DISPLAY error message if the model pose is not found
```

hw07_services_main.m

```
CLEAR variables and command window
SHUTDOWN existing ROS connections
INITIALIZE new ROS connection to specified robot simulator master and local node
INITIALIZE robot handle
STORE robot handle and configuration options
RESET robot simulation world
DEFINE gazebo as the pose retrieval method
GET all model names from the simulation world
SET loop execution speed
IF gazebo is the pose retrieval method
    FOR each model in the simulation
        GET the model name
        DISPLAY the model name
        GET the model pose
        DISPLAY the model position in xyz coordinates
        DISPLAY the model orientation as a quaternion
        WAIT to enforce the loop execution speed
```

Links:

Code: https://github.com/mrblanders/26F_matlab_ros_support/tree/main/hw/hw07_services
Video: https://youtu.be/u4vXZfOKzgQ